
CHAPTER 6

COMMON CONTINUOUS RANDOM VARIABLES

CONTINUOUS UNIFORM DISTRIBUTION

NORMAL DISTRIBUTION

STANDARD NORMAL DISTRIBUTION

APPLICATIONS OF NORMAL DISTRIBUTION

GAMMA AND EXPONENTIAL DISTRIBUTION



INTRODUCTION

- Just like with discrete RVs, some continuous RVs occur so frequently in nature that they have well-known functional forms
- We turn now to studying these distributions



CONTINUOUS UNIFORM DISTRIBUTION

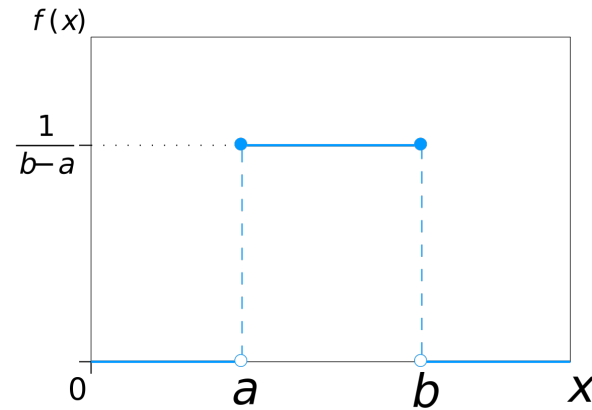


THE UNIFORM DISTRIBUTION

- A random variable X is said to have the **uniform distribution over the interval** $[a, b]$ if its pdf is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

- We say $X \sim \text{UNIF}(a, b)$.



INTEGRATE!

- Confirm this is a pdf.

$$f(x; a, b) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

CDF

- Show that the CDF is:

$$F(x; a, b) = \begin{cases} 0, & x < a \\ \frac{x - a}{b - a}, & a \leq x \leq b, \\ 1, & x > b \end{cases}$$

EXAMPLE I

- To commute a work, a professor must first get on a bus near her house. Suppose the waiting time (in minutes) at the stop follows a $\text{UNIF}(0,5)$ distribution.
 - Specify the pdf.
 - What is the probability that she has to wait at most 3 minutes?
 - What is the probability that the professor has to wait more than 4 minutes, given that she has waited at least 3 minutes?

MEAN

- If $X \sim \text{UNIF}(0,5)$, what is $E[X]$?
- What is $E[X]$ for general $X \sim \text{UNIF}(a, b)$?

MEAN AND VARIANCE IN GENERAL

- Show that $E[X] = \frac{a+b}{2}$:

- It can also be shown that $\text{Var}(X) = \frac{(b-a)^2}{12}$



EXPONENTIAL DISTRIBUTION



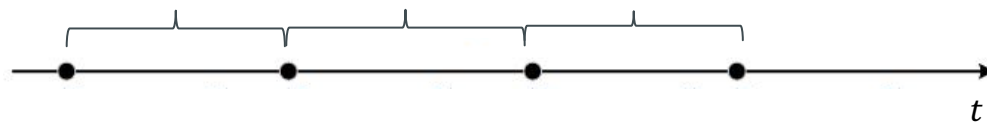
THE EXPONENTIAL DISTRIBUTION

- Motivated as time between events in a Poisson process
- Poisson process: events or arrivals occurring in space or time, with average rate of λ arrivals per unit time.

Large $\lambda \Rightarrow$ small β
distances between arrivals



Small $\lambda \Rightarrow$ large β
distances between arrives



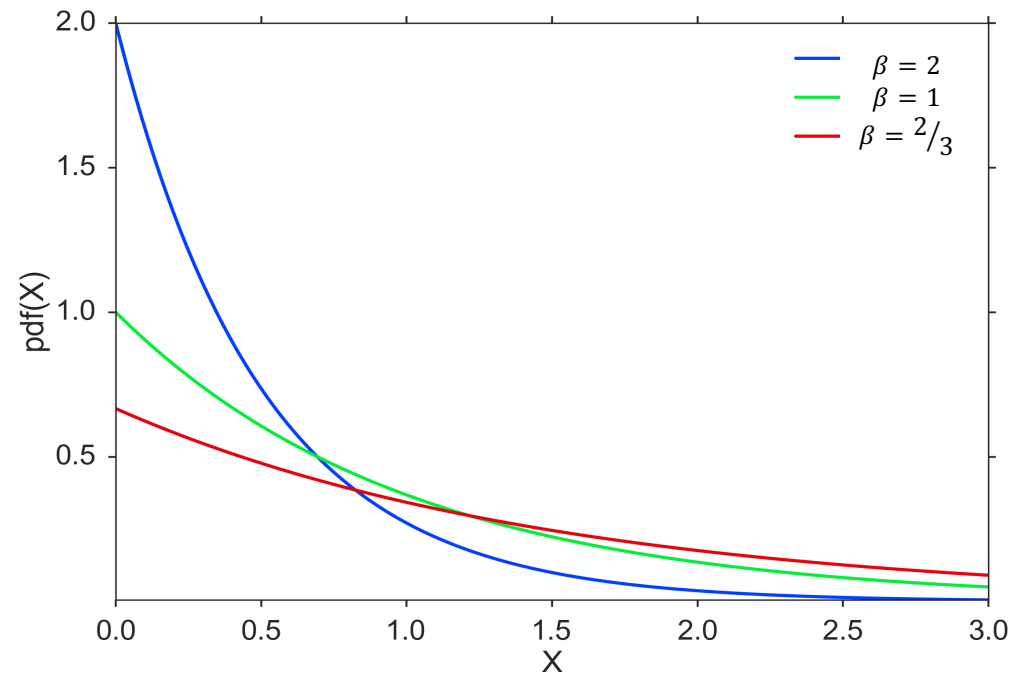
THE EXPONENTIAL DISTRIBUTION

- A random variable X is said to have an **exponential distribution with scale parameter β**
- we say $X \sim EXP(\beta)$, if its pdf is given by:

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- This pdf is defined for any $\beta > 0$

THE EXPONENTIAL DISTRIBUTION



INTEGRATE!

- Confirm this is a pdf.

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

CDF

- Show that the CDF is:

$$F(x; \beta) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x/\beta}, & x > 0 \end{cases}$$

EXAMPLE 2

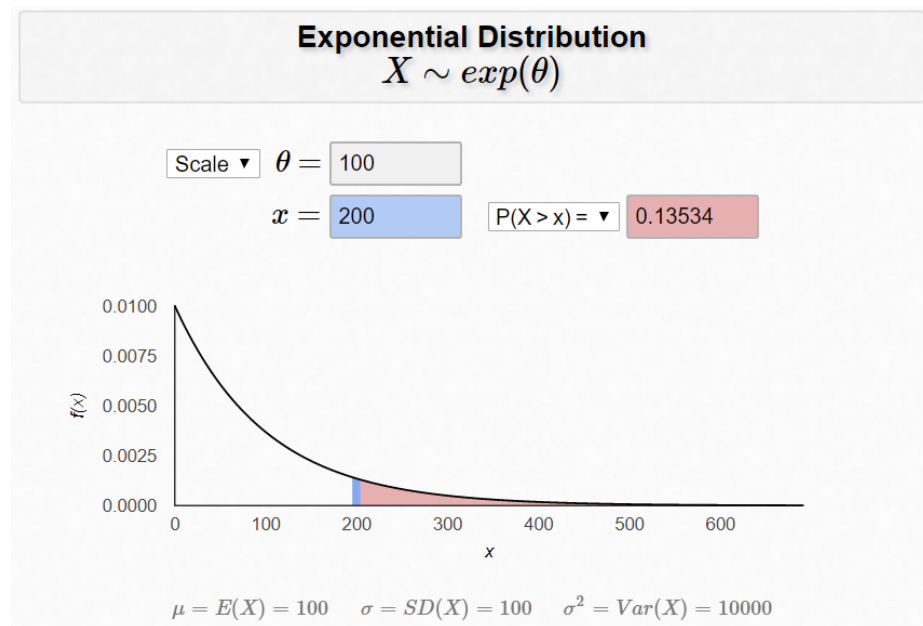
- The lifetime (in hours) of an electrical component can be modeled with an $EXP(100)$ distribution; specifically:

$$f(x) = \frac{1}{100} e^{-x/100}; x > 0$$

- What is the probability that the component will operate for at least 200 hours without failure?

TECHNOLOGY

- No calculator function.



EXAMPLE 2

- What is the probability that component lasts more than 200 minutes, given that it has lasted more than 100 minutes?

MEMORYLESS PROPERTY

- Memoryless Property: Show that in general, $P(X > x + t | X > x) = P(X > t)$

MEAN AND VARIANCE

- Show that $E(X) = \beta$

- It can also be shown that: $Var(X) = \beta^2$

ALTERNATE PARAMETERIZATION

- A random variable X is said to have an **exponential distribution with rate parameter λ**
- we say $X \sim EXP(\lambda)$, if its pdf is given by:

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- This pdf is defined for any $\lambda > 0$
- $E[X] = 1/\lambda$
- $\text{Var}(X) = 1/\lambda^2$

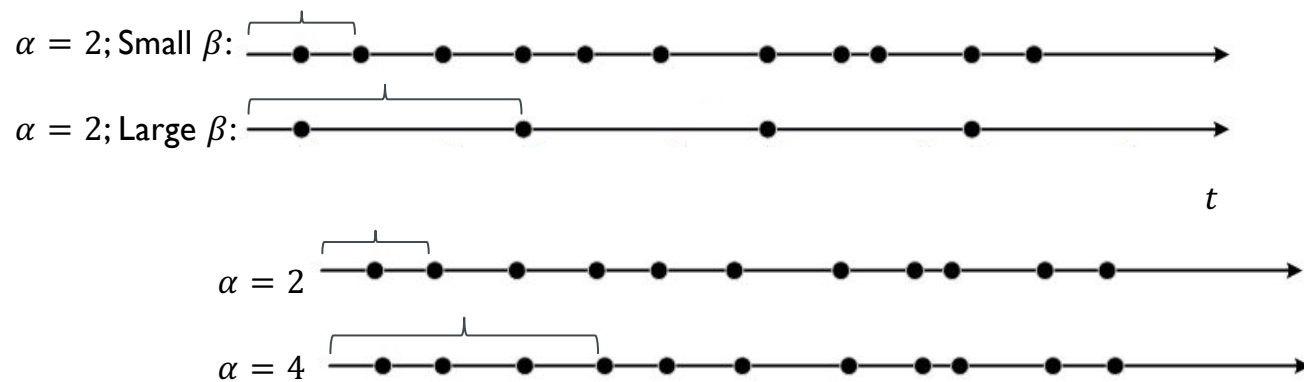


GAMMA DISTRIBUTION



THE GAMMA DISTRIBUTION

- Motivated as time until the r^{th} event in a Poisson process with arrival rate λ
- Low values of X (time until r^{th} event) are influenced by:
 - Low scale (β)
 - Small shape (α)



THE GAMMA DISTRIBUTION

- A random variable X is said to have a **gamma** distribution with scale parameter $\beta > 0$ and shape parameter $\alpha > 0$
- We say $X \sim \text{GAM}(\alpha, \beta)$, if its pdf is given by:

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- Exponential distribution is a special case of Gamma with $\alpha = 1$

THE GAMMA FUNCTION

What is that weird $\Gamma(\alpha)$? Defined as:

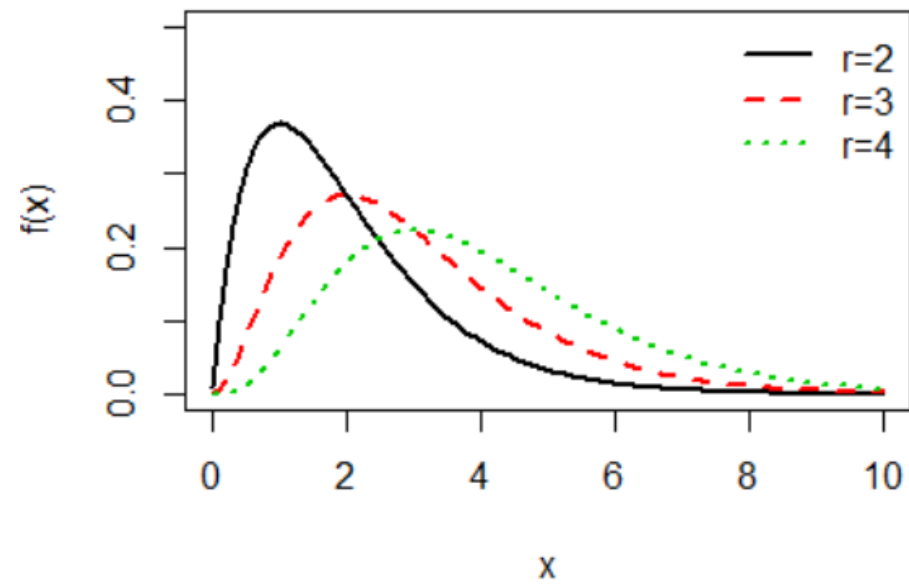
$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

Properties:

- $\Gamma(\alpha) = (\alpha - 1)!$, for any positive integer α
- $\Gamma(1) = 1$
- $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
- $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$ (integrate by parts)

THE GAMMA DISTRIBUTION

- Graphs of the pdf for various α and fixed $\beta = 1$:



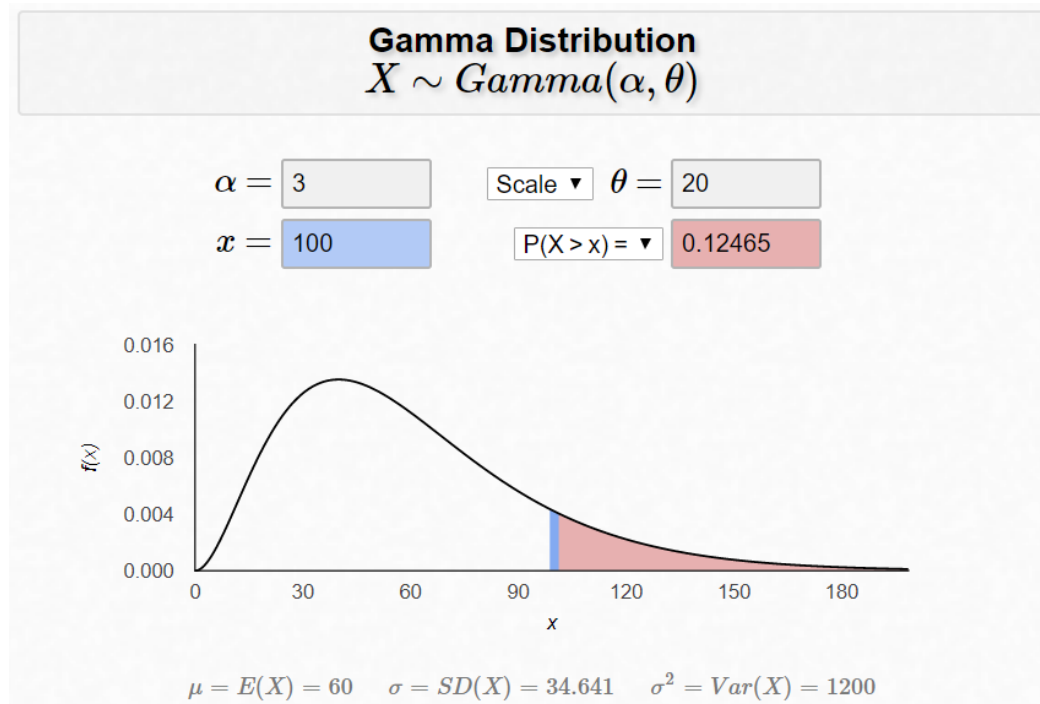
MEAN AND VARIANCE

- If $X \sim \text{Gam}(\alpha, \beta)$, then
 - $E(X) = \alpha\beta$ (waiting time increases as time between arrivals increases)
 - $\text{Var}(X) = \alpha\beta^2$

EXAMPLE 3

- Accidents at a notorious intersection follow a Poisson process. On average, an accident occurs once every 20 days. Let X denote the time (in days) you will have to wait for the 3rd accident to occur
 - What is the distribution of X ?
 - If on average, an accident occurs once every 20 days, what is the average amount of time (in days) that you expect to wait for the 3rd accident to occur?
 - If on average, an accident occurs once every 20 days, what is the probability that you will have to wait more than 100 days for the 3rd accident to occur?
 - <https://homepage.divms.uiowa.edu/~mbognar/applets/gamma.html>

TECHNOLOGY





NORMAL DISTRIBUTION





INTRODUCTION

- The most important distribution in statistics!
- Well-models many natural phenomena:
 - Height
 - Weight
 - Age

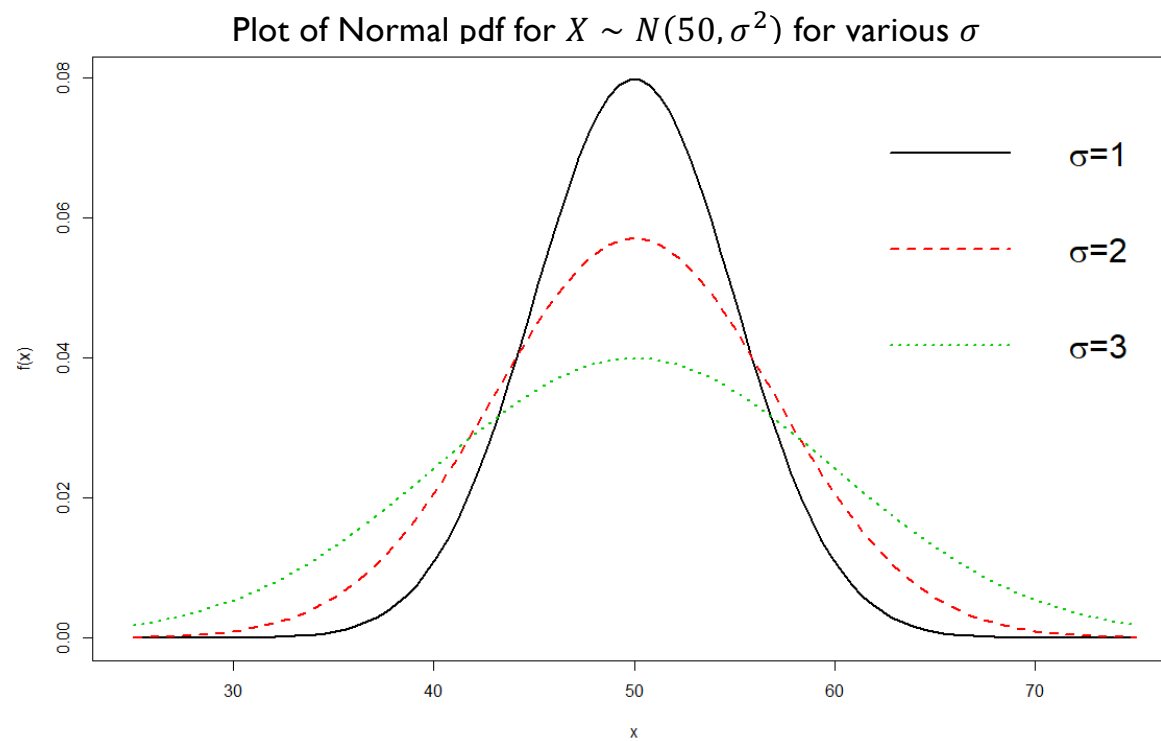
THE NORMAL DISTRIBUTION

- X is said to follow a **normal distribution** iff, for parameters μ and σ^2 with $-\infty < \mu < \infty$ and $\sigma^2 > 0$:

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty$$

- We say $X \sim N(\mu, \sigma^2)$

THE NORMAL DISTRIBUTION



MEAN AND VARIANCE

- If $X \sim N(\mu, \sigma^2)$, then
 - $E(X) = \mu$
 - $\text{Var}(X) = \sigma^2$
 - $SD(X) = \sigma$

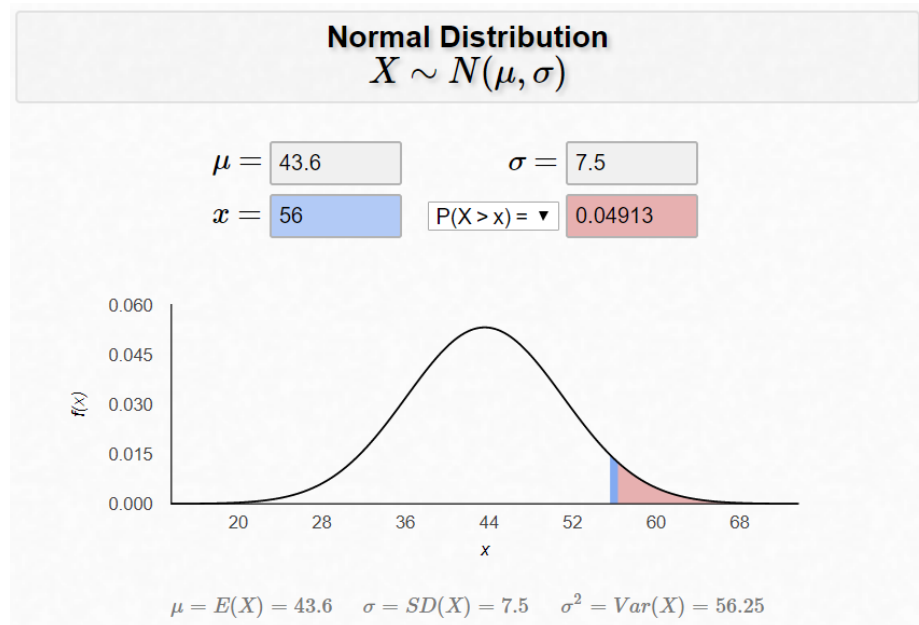
EXAMPLE 4

- In Manoa Valley on the island of Oahu (Hawaii) the annual rainfall averages 43.6 inches with standard deviation 7.5 inches and follows an approximate normal distribution. A year is chosen at random. What is the probability that the annual rainfall is more than 56 inches?

TECHNOLOGY

Calculator:

- `I-normalcdf(43.6, 56.25, 56)`



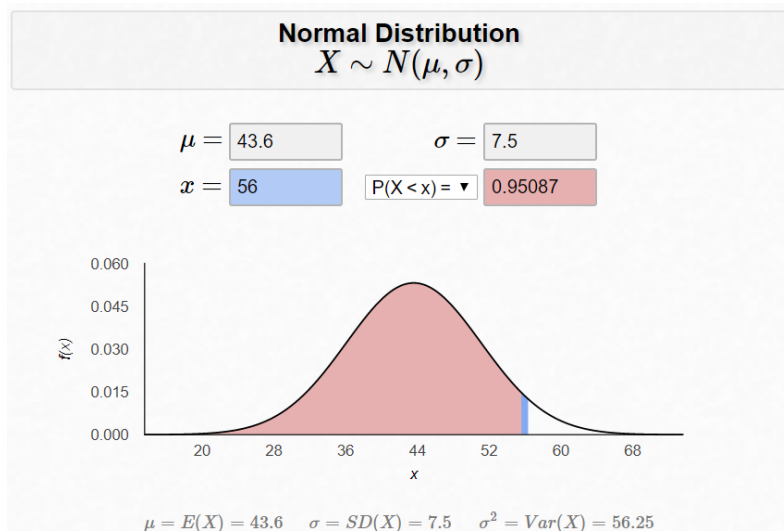
EXAMPLE 4

- In Manoa Valley on the island of Oahu (Hawaii) the annual rainfall averages 43.6 inches with standard deviation 7.5 inches and follows an approximate normal distribution. A year is chosen at random. What is the probability that the annual rainfall is between 32 and 56 inches?

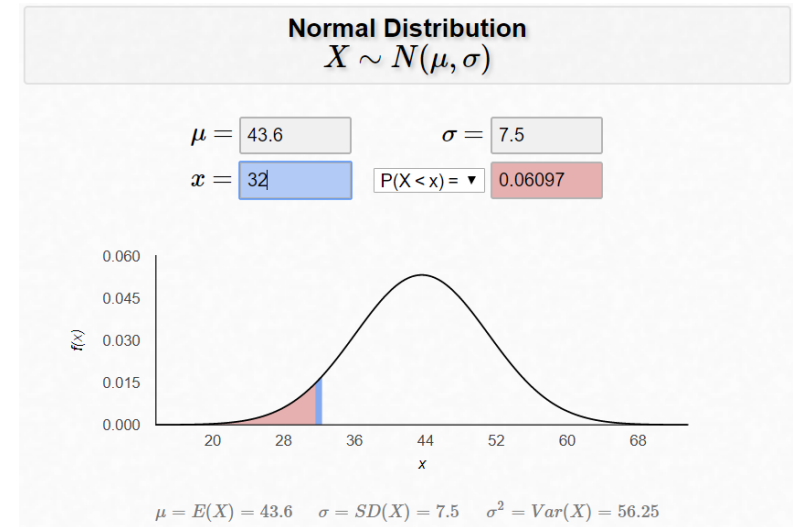
TECHNOLOGY

Calculator:

- normalcdf(43.6, 56.25, 56) – normalcdf(43.6, 56.25, 32)

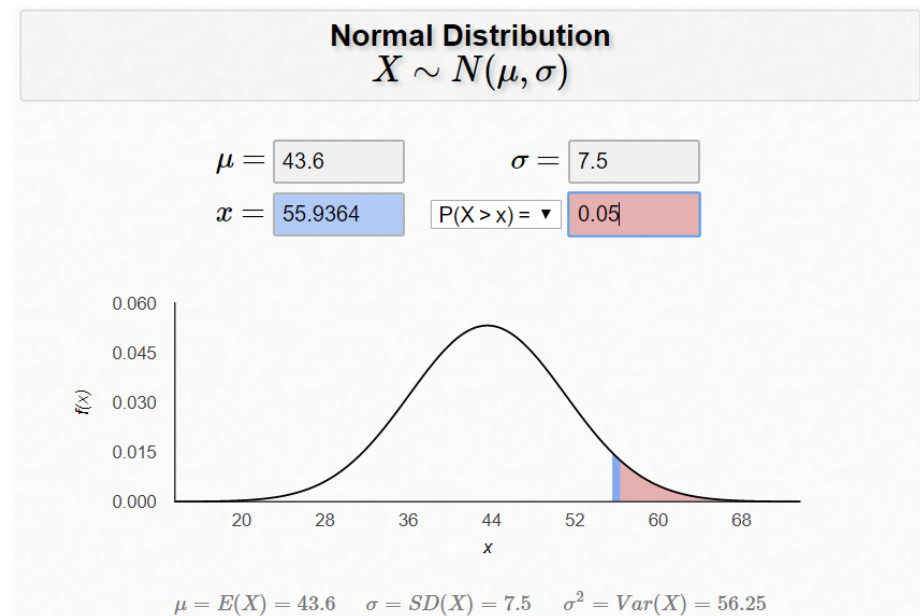


—



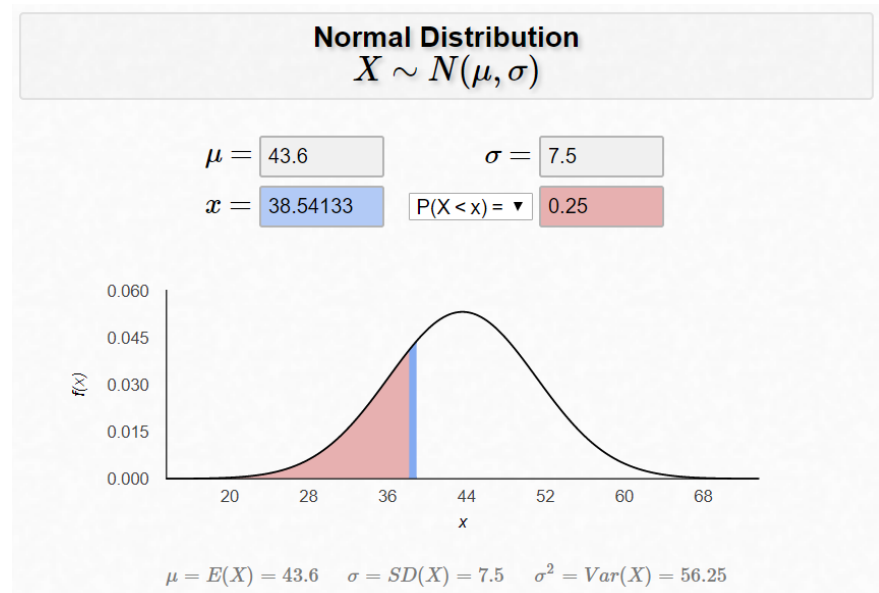
EXAMPLE 4

- In Manoa Valley on the island of Oahu (Hawaii) the annual rainfall averages 43.6 inches with standard deviation 7.5 inches and follows an approximate normal distribution. 5% of years will have rainfalls above what amount?



EXAMPLE 4

- In Manoa Valley on the island of Oahu (Hawaii) the annual rainfall averages 43.6 inches with standard deviation 7.5 inches and follows an approximate normal distribution. 25% of years will have rainfalls below what amount?



BY HAND CALCULATION

- How?
 1. Transform values (x) to z-scores
 2. Use a table to find cumulative probabilities for each z-score
 3. Find the answer

STANDARD NORMAL DISTRIBUTION

- If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1)$$

- A normal distribution with mean = 0 and variance = 1 is called the ***standard normal distribution***.

Z-SCORE

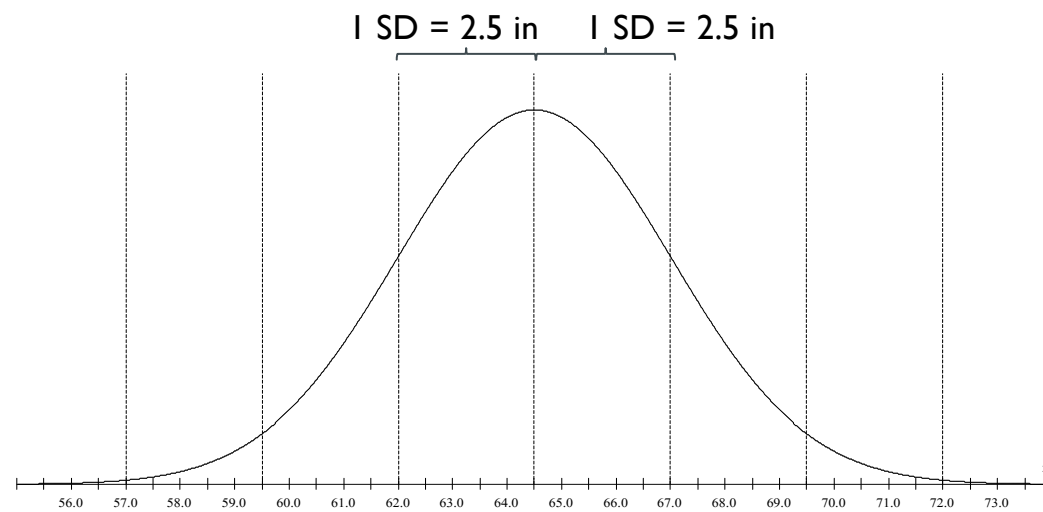
- The **z-score** of a value is given by

$$z = \frac{x - \mu}{\sigma} = \left(\frac{\text{value} - \text{mean}}{SD} \right)$$

- Measures *relative* distance to mean

EXAMPLE 5

- Suppose the heights of women are Normally distributed with mean = 64.5 in. and S.D. = 2.5 in.:



Read z

Table A.3 Areas under the Normal Curve



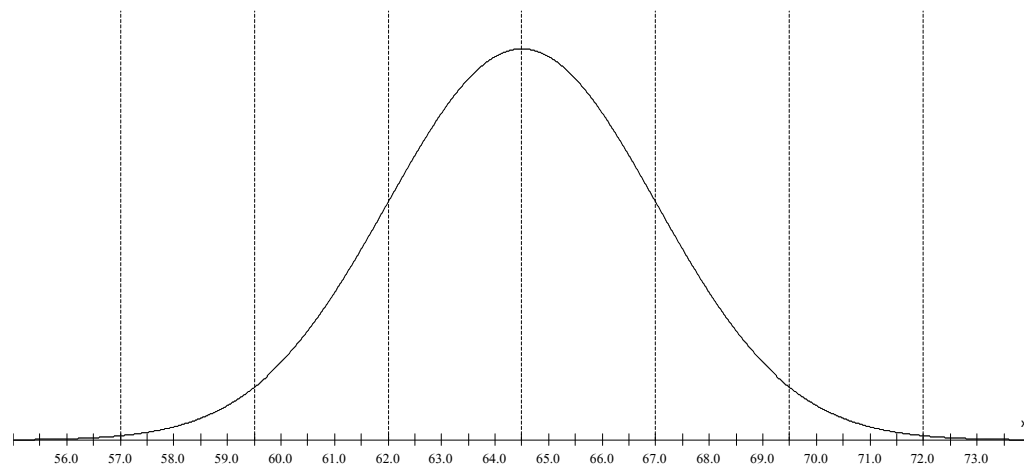
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

TECHNIQUE (AGAIN)

- Write probability in terms of X
- Convert to probability in terms of z-score
- Use table

EXAMPLE 5: SHADING THE PROBABILITY

- What is the probability a young woman is taller than 60 inches?
 - Shade the corresponding area.

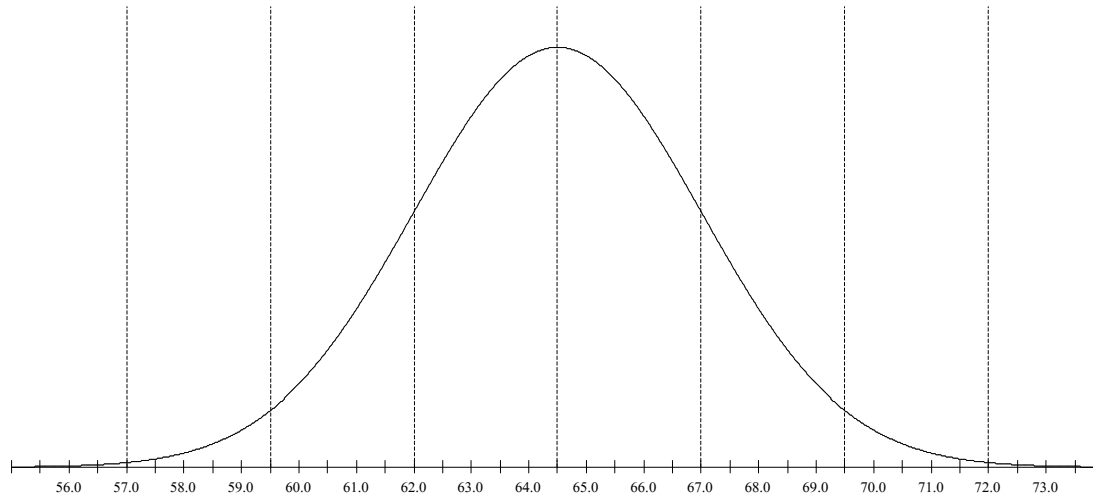


EXAMPLE 5: USE TABLE

- Suppose the heights of women are Normally distributed with mean = 64.5 in. and S.D. = 2.5 in. What is the approximate probability a woman is taller than 60 inches?

EXAMPLE 5: SHADING THE PROBABILITY

- What is the probability a young woman is between 61 and 67 inches tall?
 - Shade the corresponding area.

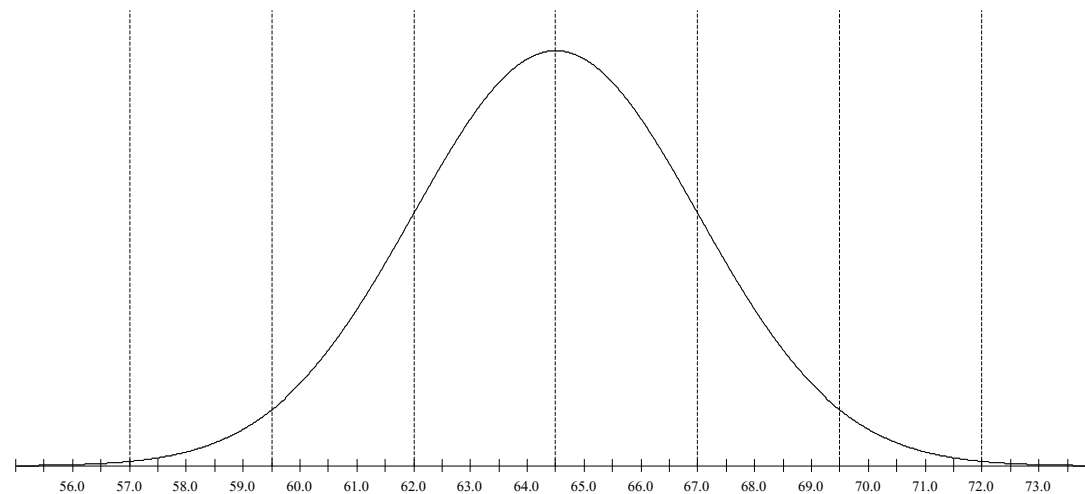


EXAMPLE 5: USE TABLE

- Suppose the heights of women are Normally distributed with mean = 64.5 in. and S.D. = 2.5 in. What is the approximate probability a woman is between 61 and 67 inches?

EXAMPLE 5: SHADING THE PROBABILITY

- What is the probability a young woman is shorter than 63 inches tall?
 - Shade the corresponding area.



EXAMPLE 5: USE TABLE

- Suppose the heights of women are Normally distributed with mean = 64.5 in. and S.D. = 2.5 in. What is the approximate probability a woman is shorter than 63 inches?



CHI – SQUARE DISTRIBUTION



CHI-SQUARE DISTRIBUTION

- A continuous random variable X is said to follow a **chi-square distribution**.
- We say χ^2_ν with pdf

$$f(x; \nu) = \frac{1}{2^{(\frac{\nu}{2})} \Gamma(\nu/2)} x^{(\nu/2)} e^{-x/2}, x > 0$$

- Where $\nu > 0$
- Special case of the gamma distribution with GAM $\left(\alpha = \frac{\nu}{2}, \beta = 2\right)$
- If $X \sim \chi^2_\nu$, then $E[X] = \nu$ and $\text{Var}(X) = 2\nu$

CHI-SQUARE PROPERTIES

- Recall sample variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{\sum_{i=1}^n x_i^2 - n \bar{x}^2}{n - 1}$$

- Most importantly, the sample variance s^2 is related to a χ_{n-1}^2 distribution
 - We will talk about this more in chapter 8.